Zhiyu Wu

zhiyuwu2@illinois.edu • **https://xzzwzy.github.io/**

Education

University of Illinois Urbana-Champaign — **Champaign, IL**
Ph.D. Computer Science — *Aug 2026 –present*
Advisor: Prof. Fan Lai
Research Interest: Systems for ML, Physical AI Systems, Inference/Training/RL Software stacks, Efficient ML

University of Illinois Urbana-Champaign — **Champaign, IL**
M.S. Computer Science – Thesis Based — *Aug 2024 – Dec 2025*

University of Michigan — **Ann Arbor, MI**
B.S.E. Computer Engineering (Dual Degree) — *Aug 2022 – May 2024*

Shanghai Jiao Tong University — **Shanghai, China**
B.S.E. Electrical and Computer Engineering (Dual Degree) — *Sept 2020 – Aug 2024*

Publications

- JITServe: SLO-aware LLM Serving with Imprecise Request Information; NSDI, 2026; Wei Zhang*, **Zhiyu Wu***, Yi Mu, Rui Ning, Banruo Liu, Nikhil Sarada, Myungjin Lee, Fan Lai
- CacheFlow: Efficient LLM Serving with 3D-Parallel KV Cache Restoration; Preprint, 2026; Sean Nian, Jiahao Fang, Qilong Feng, **Zhiyu Wu**, Fan Lai
- Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services; Preprint, 2024; Jiachen Liu, **Zhiyu Wu**, Jae-Won Chung, Fan Lai, Myungjin Lee, Mosharaf Chowdhury
- The ML. ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization; NeurIPS, spotlight, 2025; Jae-Won Chung, Jiachen Liu, Jeff J Ma, Ruofan Wu, Oh Jun Kweon, Yuxuan Xia, **Zhiyu Wu**, Mosharaf Chowdhury

Work Experience

Software Engineering on MLsys, Bytedance, AML — **San Jose, CA**
Supervisor: Xiao Yu, Zilin Xu — *Jan 2026 –Aug 2026*

- Optimize training performance of Seed model series on VeOmni (default trainer backend of VeRL) and VeRL on B200/GB200, including framework and kernel optimizations.
- Work on Post-training pipeline and data plane of LLM training.
- Establish TPU inference stack with colleagues and collaborators from Google, especially, working on KV cache management, DP/TP, MTP adaption, and adaption to Seed models.

Research Engineering Intern, Bytedance, AML — **San Jose, CA**
Supervisor: Xiao Yu — *May 2025 –Aug 2025*

- Built end-to-end AI model training workflow with Jupyter notebook and web frontend, encompassing dataset profiling, training parameter optimization, SFT/RL deployment, evaluation, and production inference pipeline.
- Contributed to LLM training simulator with memory analysis, communication modeling, and MFU metrics.
- Explore and optimize memory usage for SFT/pretrain with FSDP, specifically, optimize compiler/kernel to maximum tensor reuse in backward and optimizer stage.

Research Experience

Sentry: Proactive Agents Execution with State Lineage — **Champaign, IL**
Supervisor: Fan Lai — *Oct 2025 - present*

- Designed a lineage-based control plane for stateful LLM agents, combining cost-aware checkpoint loading and replay to selectively recover from errors detected late while preserving independent work.
- Combined execution-graph matching with periodic tool-call speculation to prepare state and execute tools during model reasoning, validating action equivalence and input-state versions before reusing results.

- Designed demand-driven KV cache write-back and GPU-memory lending during tool stalls, with prediction-guided prefetch and page reclamation to admit new requests and restore paused contexts.
- Designed guarded asynchronous execution and sandbox colocation using learned dependency contracts, private state versions, and ordered commits, with conflict-triggered recovery and workspace splitting.

CacheFlow: Efficient LLM Serving via Automated 3D-Parallel KV Cache Restoration **Champaign, IL**
Supervisor: Fan Lai *Aug 2024 – Sept 2026*

- Formulated KV cache restoration as a multidimensional scheduling problem, coordinating recomputation and cache loading across tokens, layers, and GPUs.
- Developed a batch-aware dynamic-programming planner that jointly selects recomputation boundaries across requests and layers, accounting for heterogeneous full-attention and recurrent-layer costs.
- Enabled concurrent restoration across model shards by caching lightweight boundary activations, removing sequential recomputation dependencies between GPUs.
- Implemented CacheFlow atop vLLM and LMCache, reducing TTFT by 10%–62% across diverse models, chatbot and agentic workloads, and hardware configurations.

JITServe: SLO-aware LLM Serving with Imprecise Request Information **Champaign, IL**
Supervisor: Fan Lai *Jul 2024 – Sept 2025*

- Developed an SLO-aware scheduler for latency-sensitive, throughput-sensitive, and compound requests, allocating resources to meet heterogeneous performance requirements.
- Represented multi-agent workflows as DAGs and applied graph matching to predict execution dependencies and remaining workload; used quantile regression forests to estimate response lengths.
- Designed service-density-based prioritization to maximize service gain while preserving capacity for other requests and preventing starvation of lower-priority workloads.
- Achieved 1.3–8.3× higher service gain and 4.0–10.3× higher SLO goodput than existing serving systems across diverse workloads and models.

Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services **Ann Arbor, MI**
Supervisor: Mosharaf Chowdhury, Amber Liu *May 2023 – April 2024*

- Identified the mismatch between conventional serving metrics and streaming user experience, highlighting the importance of delivering text at a pace that matches user reading speed.
- Defined a Quality-of-Experience (QoE) metric that tracks token delivery throughout generation and captures its impact on the user's reading experience.
- Formulated online scheduling as a knapsack optimization problem, allocating serving resources to maximize overall QoE across concurrent requests.
- Built Andes atop vLLM and integrated the QoE-aware scheduler into the serving runtime to improve the responsiveness of LLM text streaming.

Project Experience

Triton GPU Programming and Attention Kernels **San Jose, CA**
Developer *April 2026 – July 2026*

- Extend the official Triton tutorials into a structured learning resource, adding detailed explanations and implementation walkthroughs for GPU kernel programming.
- Developed in-depth tutorials on FlashAttention and other attention variants, connecting algorithmic design with Triton kernel implementations and performance considerations.
- Co-developed and delivered an eight-session internal lecture series for the ByteDance AML group, covering Triton fundamentals and attention kernel implementation

Symbiotic Lab/ML.ENERGY.LEADERBOARD Team **Ann Arbor, MI**
Developer *May 2023 – Sept 2023*

- Developed the ML.ENERGY Leaderboard, an open-source platform for benchmarking the energy efficiency and NLP performance of LLM models.

- Defined performance metrics and implemented scripts for optimized batched inference to ensure accurate measurement.
- Contributed to the online Chatbot Arena which gathers data on models' energy consumption and performance.

Professional Service

- VP 160 Honors Physics, SJTU *Summer 2022*
- Volunteer teacher for Mine Village Primary School, Luxi Town, Yunnan Province, China. *Dec 2021 – Jan 2022*

Honors

Dean List, *Umich* 2023
University Honor, *Umich* *2022*
Tang Junyuan Scholarship, *SJTU* *2022*
SJTU Undergraduate Excellent Scholarship Class B, *SJTU* *2022*